Responsible Multi-Omic Neural Architecture Search: Ethical, Social, and Public-Health Considerations in the Spatio-Temporal Bio-NAS Program

Breast cancer anchor cohort, controlled-access neurodegenerative data, and open reproducibility via Docker

Authoring Note

This white paper was drafted and edited with assistance from **Cursor Composer 2.5** *(AI). The human author reviewed all content, verified project-specific facts against the Bio-NAS repository and master plan.*

INTRODUCTION AND OBJECTIVES

Motivation and intellectual merit

The Spatio-Temporal Dual-Track **Bio-NAS** program asks whether biologically informed neural architecture search—where connectivity is constrained by curated pathway knowledge (KEGG, Reactome) via masked linear layers—outperforms unconstrained neural architecture search on multi-omic disease trajectory prediction. The thesis design runs a rigorous **Track A (control)** versus **Track B (Bio-NAS)** comparison on a breast invasive carcinoma (BRCA) anchor cohort before scaling, under a documented **scaling gate**, to Alzheimer's disease, rheumatoid arthritis, type 2 diabetes, and epigenetic aging cohorts. Intellectual merit lies in coupling spatio-temporal deep learning (genomic "spatial" axes and longitudinal Δt embeddings) with reproducible experiment orchestration (Optuna, patient-level holdout, causal cross-validation).

Broader impacts and objectives

Multi-omic machine learning touches sensitive human data, uneven cohort representation, and pathways toward clinical decision support. Even when the immediate goal is research software, stakeholders may misread pipeline smoke metrics as diagnostic evidence, re-identify participants from rich omics tables, or deploy models trained on convenience cohorts (for example TCGA) as if they were population-representative. This white paper maps **concrete Bio-NAS design choices** to ethical, social, environmental, and economic implications, and pairs each risk with mitigations already planned or implemented in the repository.

Objectives of this document are to: (1) explain why Bio-NAS raises distinct governance challenges compared with tabular clinical models; (2) analyze several project-native examples—mislabeled "toy" metrics, controlled-access data handling, and equity in cohort composition, omics-linked re-identification risk, and interpretability claims; and (3) align the three-year technical roadmap with responsible outcomes. In the same spirit of transparency applied to model outputs, we note that **Composer 2.5 assisted in drafting this paper**, while factual claims about Bio-NAS were checked against project documentation (for example docker/README.md, docs/data/cohort_inventory.md, and phd_bio_nas_master_plan.md).

Non-goals

Plan-1 Docker demonstrations download small **open-access** public slices (for example TCGA-BRCA Level-3 methylation and RNA) and run CPU toy architecture search. Project documentation states explicitly that these runs **validate engineering workflows only** and are **not clinical evidence**. Full holdout benchmarks and attribution interfaces are Year 2–3 deliverables; until then, AUROC-style numbers from smoke demos must not inform patient care.

Issue 1: Toy metrics mistaken for clinical truth

The epigenetic aging Docker smoke uses chronological age thresholds (for example age ≥ 50 as a binary label) solely to exercise the training pipeline; wiki text labels this a **pipeline validation label**, not clinical aging acceleration or disease status. Similarly, the static HTML dashboard and train_toy_nas_cpu.py distinguish clinical proxies (such as AJCC staging on BRCA smoke) from **synthetic** targets on other diseases. The dashboard builder embeds an explicit HTML notice that toy metrics **are not clinical results** (scripts/build_dashboard.py). BRCA toy training uses leave-one-out cross-validation on roughly five to ten joint methylation cases—variance is enormous, yet a single AUROC figure can look authoritative in a slide deck. **Implication:** Without persistent labeling discipline, funders or collaborators may over-trust metrics from tiny public slices and omit uncertainty when communicating public-health relevance. **Mitigation:** Frozen nomenclature ("smoke" versus "holdout"), dashboard disclaimers, reporting confidence intervals or bootstrap bands on holdout work (Year 2), and planned OSS release notes (Summer 2029) that separate demo artifacts from frozen experiment IDs cited in the dissertation.

Issue 2: Data governance, privacy, and reproducibility

Alzheimer's work locks **ADNI (LONI)** as the primary cohort, which requires an approved Data Use Agreement; the Docker entrypoint **scaffolds and skips** controlled download when credentials are absent—no automated LONI scrape—and stores skip status under data/adni/. Credentials belong in a gitignored host .env, never in the image. Type 2 diabetes primary **KORA** and dbGaP-controlled alternates (for example AURORA metastatic breast resources) remain **outside Docker** until project agreements complete, while the open reproducibility path documents honest limits in cohort_inventory.md. Optional pipeline automation (GitHub Repository Dispatch after toy runs) is **disabled by default** in Compose documentation so CI webhooks cannot become an exfiltration path for paths, tokens, or cohort filenames. **Implication:** Open science and privacy pull in opposite directions if reproducibility is misread as "all cohorts ship in Compose." **Mitigation:** Inventory verification lists controlled sources under skipped_controlled; human-in-the-loop DUA steps are documented exceptions to Docker-first implementation; post-DUA verification only after approval; no secrets baked into images or dispatch payloads.

Issue 3: Equity, representation, and multi-cohort fusion

TCGA-BRCA (~1,098 cases in inventory verification) is a practical anchor but **not demographically representative** of all patients with breast cancer; true repeated molecular timepoints are **sparse** on TCGA, with AURORA US documented as an alternate when paired primary–metastasis omics matter. Epigenetic aging harmonization (GSE40279, GSE87571, GSE280465) must address **tissue-specific and age distribution biases** per master-plan acceptance criteria. Rheumatoid arthritis integrates **unpaired** cohorts (GSE71841 methylation versus DAS28 severity anchors) with multi-task loss masking so severity is not falsely trained on the small healthy/RA methylation set (n≈24). **Implication:** Models may absorb ancestry, sex, or batch effects as apparent biology. **Mitigation:** Patient-level 80/20 holdout with automated leakage tests; subgroup sensitivity analyses in thesis synthesis; **scaling gate** limiting multi-disease Track B claims unless BRCA holdout shows advantage on accuracy, interpretability, and/or sparsity.

Issue 4: Re-identification, omics richness, and public-health privacy

Multi-omic tensors combine hundreds of thousands of CpG or gene features with visit timestamps and clinical covariates. Even when identifiers are removed from public releases, high-dimensional profiles can support **re-identification** or kin inference when linked with auxiliary data [19]—a concern for TCGA Level-3 open tables used in Plan-1 Docker smoke (~5–10 joint BRCA cases) and, under DUAs, for ADNI blood EPIC methylation with repeated clinical waves. The welfare issue is not only confidentiality breach but loss of public trust in legitimate research reuse. **Implication:** Static dashboards, Streamlit demos, and attribution exports can become accidental disclosure channels if they ship raw case IDs, full beta matrices, or downloadable holdout rows without access control. **Mitigation:** Keep controlled cohorts off public Compose mounts; use de-identified demo identifiers in any Year 3 UI; export aggregate or top-k attribution summaries for OSS; document auth and PII constraints as a master-plan acceptance item before demos that resemble clinical workflow; align ADNI use with LONI DUA terms after approval.

Issue 5: Interpretability, pathway priors, and welfare of explanation

Track B constrains connectivity using KEGG and Reactome adjacency translated into **MaskedLinear** layers, aiming for sparsity and biologist-readable structure compared with Track A's unconstrained search. Year 3 Captum/SHAP pipelines will rank CpG sites and timepoints driving predictions, with Track B maps checked against pathway priors. Explanations can improve **human oversight**, but they are not causal proof: pathway catalogs are incomplete, historically Eurocentric in curation, and SHAP attributions in genomics can be unstable under preprocessing shifts. Treating pretty heatmaps as mechanistic certainty risks **interpretability washing**—especially if Streamlit curves are shown to patients or families without context. **Implication:** Overconfident mechanistic narratives may harm welfare by steering attention toward wrong targets or delaying appropriate care. **Mitigation:** Label visualization outputs as **research hypotheses** requiring clinician and statistician review; pair attributions with holdout performance and ablations; use the **scaling gate** so multi-disease "Bio-NAS wins" claims wait on BRCA evidence for accuracy, interpretability, and/or sparsity.

Global, cultural, and social context

Bio-NAS spans oncologic, neurologic, autoimmune, metabolic, and aging-related cohorts drawn from **different continents and access regimes**. TCGA-BRCA is a US-heavy convenience anchor; KORA T2D is a European controlled resource; public GEO smokes for RA and epigenetic aging are small series useful for engineering but weak for global generalization. KEGG/Reactome priors embed a **curated Western biomedical canon** that may under-represent pathways

shaped by ancestry, diet, or environmental exposure. Epigenetic clock harmonization (GSE40279, GSE87571, GSE280465) intersects social debates about "biological age" versus chronological age and fairness in insurance or employment—motivating the master-plan requirement to address tissue-specific and age-distribution biases before severity tiers drive multi-task losses. **Implication:** Findings may read as universal when they reflect sampling and catalog bias. **Mitigation:** Document alternates (AURORA, ROSMAP) in the inventory; plan subgroup sensitivity analyses in thesis synthesis; report null or negative **scaling gate** outcomes as valid scientific contributions that narrow public claims.

Environmental and economic factors

Year 2 distributed Optuna search on Slurm carries non-trivial **energy use and institutional or cloud cost**, borne by universities or grant budgets rather than patients directly—yet carbon and compute spending are public-good questions for funders. Track B explicitly targets **computational sparsity** via pathway constraints; if sparsity improves without sacrificing holdout performance, that outcome supports both interpretability and greener large-scale search. **Mitigation:** Log trial counts and worker scale-out in experiment metadata; report sparsity and accuracy jointly; prefer pathway-constrained models when they reduce parameter count without harming holdout metrics.

Recommended safeguards summary

The following actions map the issues above to accountable practice throughout the PhD timeline: (1) **Labeling discipline**—never present smoke or toy AUROC as clinical performance; keep dashboard and README disclaimers visible. (2) **Controlled data hygiene**—DUAs and KORA agreements outside Docker; credentials only in gitignored .env; verify ADNI only post-approval. (3) **Methodological fairness**—patient-level holdout before preprocessing, leakage tests, and planned subgroup analyses. (4) **Responsible scaling**—honor the BRCA **scaling gate** before broad Track B claims across Other 4 diseases. (5) **Safe interfaces**—Streamlit auth/PII documentation before trajectory demos; attribution as hypothesis, not diagnosis. (6) **OSS boundaries**—tagged release with DATA_USE.md, no patient identifiers in public artifacts, and optional automation off by default.

PROJECT PLAN

Technical plan

The technical plan follows the three-year Bio-NAS master roadmap, with ethics and welfare coupling at each phase.

Year 1 (Spring 2026 – Spring 2027) — sourcing, tensors, infrastructure. Spring 2026 completes the five-disease **cohort inventory** with license and ethics columns (docs/data/cohort_inventory.md). Winter 2026–Spring 2027 builds leakage-safe four-dimensional tensors (batch, time, spatial features, channels), Δt embeddings for irregular visits, and Dockerized PostgreSQL for Optuna metadata. Ethics coupling: document DUAs and IRB-aware ADNI use; defer controlled dbGaP files; enforce patient-level holdout before preprocessing to reduce optimistic bias that could later harm translation efforts.

Year 2 (Summer 2027 – Spring 2028) — dual-track NAS and benchmarking. Modular spatial (1D-CNN, spatial transformer) and temporal (ConvLSTM/GRU, attention) blocks feed separate Track A and Track B Optuna studies on BRCA. Winter 2027 runs large-scale search with causal cross-validation; Spring 2028 evaluates frozen **20% holdout** trajectories, ablations, and classical baselines—the **scaling gate** evidence base. Ethics coupling: report Track A versus Track B deltas with uncertainty; if Track B does not improve interpretability or sparsity, narrow claims before Other 4 scaling.

Year 3 (Summer 2028 – Summer 2029) — interpretability, interface, release. Summer 2028 implements Captum/SHAP attribution over CpG sites and timestamps, with Track B maps checked against pathway priors. Fall 2028 delivers a Streamlit trajectory dashboard; master-plan acceptance requires **auth and PII constraints documented** before any demo resembling clinical workflow. Winter 2028–Summer 2029 synthesizes cross-disease taxonomy (post–scaling gate) and releases tagged open-source Python with install docs and smoke boundaries.

Anticipated outcomes

Scientific outcomes: Quantitative evidence on whether KEGG/Reactome-constrained MaskedLinear Bio-NAS beats unconstrained NAS on BRCA holdout for prediction, interpretability (pathway-plausible attributions), and sparsity. Societal outcomes: A reproducible Docker path for public data only; transparent status files for controlled cohorts; attribution exports intended for expert human review, not autonomous triage. Risk monitoring: Over-interpretation of

SHAP/Captum maps as causal mechanisms; performance disparities across subgroups; accidental credential commits in OSS forks—each tracked via inventory honesty, holdout discipline, and planned DATA_USE.md at release.

Schedule

Period	Technical focus	Ethics and governance milestones
Y1 Spring 2026	All-5 inventory; BRCA PoC readiness	License/ethics table; ADNI DUA application (in progress)
Y1 Summer 2026 – Spring 2027	Feature maps; 4D ETL; Optuna hub	Leakage tests; controlled cohorts marked outside Docker
Y2 Summer 2027 – Spring 2028	Track A/B NAS; holdout benchmarks	Scaling gate decision; causal CV documented
Y3 Summer – Fall 2028	Attribution; Streamlit dashboard	PII/auth review; demo ≠ clinical product
Y3 Winter 2028 – Summer 2029	Taxonomy; thesis; OSS tag	Release notes; smoke vs holdout nomenclature frozen

Calendar authority: Bio-NAS README and phd_bio_nas_master_plan.md (Spring 2026 through Summer 2029).

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